KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2022
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

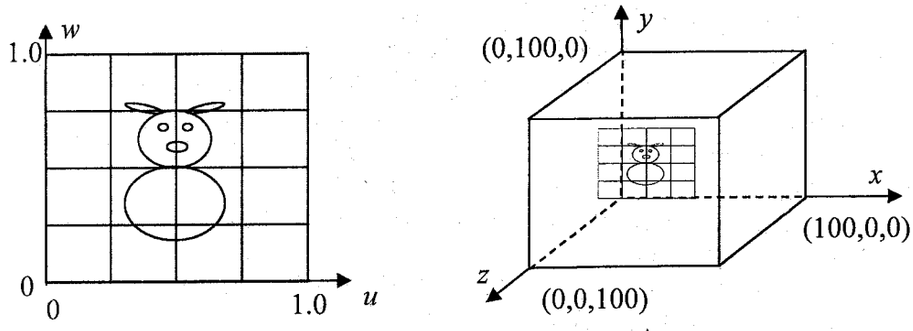
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

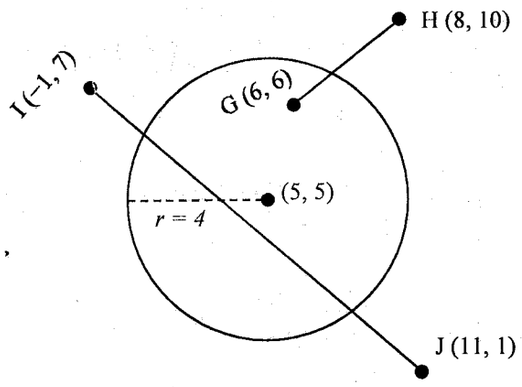
(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07) [CO1]

b) Consider the following figure and find the linear functions that map the normalized image onto a square area of 50 × 50 in the middle of the front face of the cubic object. (12) [CO1]

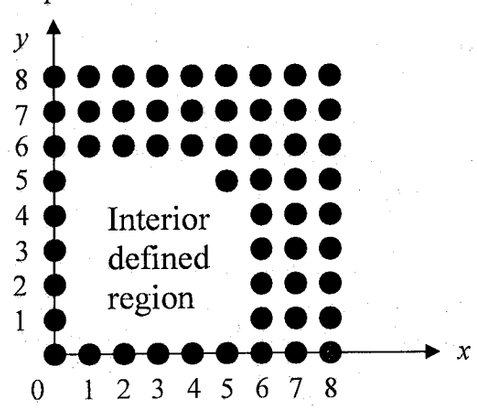


c) Use the Liang-Barsky algorithm to clip the lines in following figure. The clipping window is an axis aligned square that tightly encompasses the given circle. (16) [CO2]



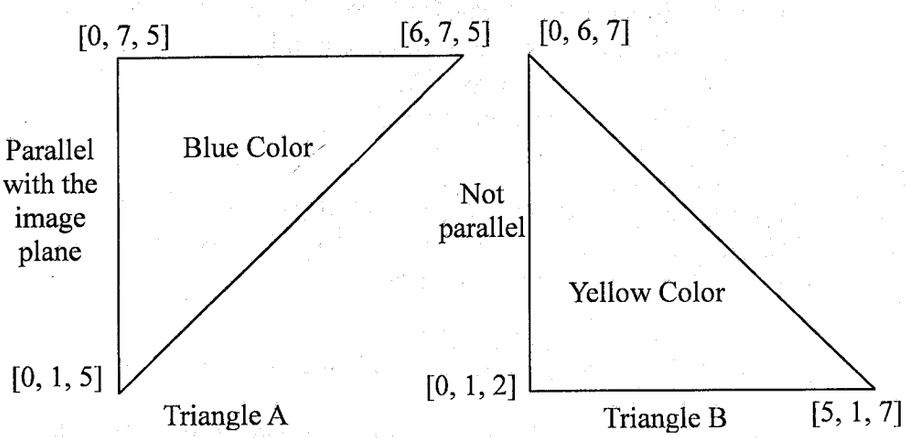
2. a) Indicate which raster locations would be chosen by Bresenham's algorithm when scan converting a line from point (1, 9) to (6, 2). (12) [CO1]

b) How would a flood-fill algorithm fill the region shown in following figure, using the 8-connected definition for region pixels? (13) [CO3]



c) When eight-way symmetry is used to obtain a full circle from pixel coordinates generated for the 0° to 45° or the 90° to 45° octant, certain pixels are set or plotted twice. This phenomenon is sometimes referred to as overstrike. Identify where overstrike occurs. (10) [CO1]

3. a) Using a 8×8 display, show how the z-buffer algorithm would determine the color of each pixel for the given triangles A (Blue Color) and B (Yellow Color) in following figure. Write 'B' for blue color and 'Y' for yellow color in the color buffer. (15) [CO2]



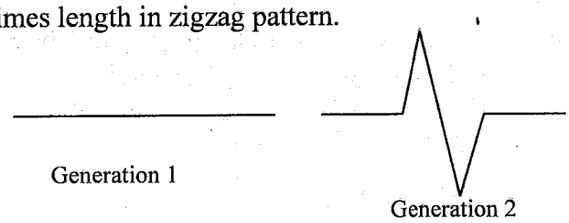
b) Explain Sutherland-Hodgeman polygon clipping algorithm. Mention it's limitations. (12) [CO3]

c) Briefly discuss how the roughness of a surface can be simulated using texture mapping. (08) [CO1]

4. a) Let, $Q(t) = (1 + t^2)p_1 + (5t - 5t^2)p_2 + t^2p_3$, and $R(t) = (1 + t^2)^2p_3 + (5t - 5t^2)p_4 + t^2p_5$ be the parametric equations of two curves, where t is the parameter and P_1, P_2, P_3, P_4, P_5 are the points on the curves. Determine whether $Q(t)$ and $R(t)$ are C^0, C^1, and G^2 continuous. (07) [CO1]

b) A Bezier curve is to be drawn given the control points $P_1(0,0)$, $P_2(3,3)$, $P_3(8,1)$, $P_4(10,2)$. Calculate the coordinates of the points on the curve corresponding to the parameter $t = 0.2, 0.4, 0.5, 0.6, 0.8$. Draw a rough sketch of the curve and show coordinates of the various points on it. (16) [CO3]

c) Two generation of a fractal is shown below, where middle one third of original curve is replaced with 4-times length in zigzag pattern. (12) [CO3]



i) Draw the third generation,
ii) Determine length of n-th generation of the fractal,
iii) Determine the dimension of the fractal.

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Find the composite transformation matrix (affine matrix) that transforms the triangle with vertices $A(1,2)$, $B(5,2)$, and $C(3,5)$ into vertices $A'(-1,-2)$, $B'(-4,-5)$, $C'(2,-2)$ respectively. (14) [CO3]

b) Determine the reflection of a 2D point $P(5,9)$ about a line $L(t) = (2,11) + (3,9)t$. Let you are given with the following reflection transformation matrix: (12) [CO2]

$$M_L = \frac{1}{m^2+1}\begin{bmatrix} 1-m^2 & 2m & -2bm \\ 2m & m^2-1 & 2b \\ 0 & 0 & m^2+1 \end{bmatrix}$$

c) Show that the path of a ray transmitted through a surface is $\mathbf{t} = \mathbf{z} - (\sqrt{1 - |\mathbf{z}|^2})\mathbf{n}$, where $\mathbf{z} = \frac{\eta_1}{\eta_2}(\mathbf{u} - (\mathbf{u}.\mathbf{n}))\mathbf{n}$, $\mathbf{u}$ is an incoming ray, $\mathbf{n}$ is the surface normal, and η_1, η_2 are the refractive indices of the media. (09) [CO3]

6. a) The following matrix is a general parallel projection matrix with direction of projection (DOP) onto an arbitrary plane with plane normal N, passing through P_0. Now, determine the parallel projection of the point $(2,3,4)$ if DOP is $(1,1,1)$ and projection plane is $2x + y + z = 8$. Symbols have their usual meanings in the given matrix. (12) [CO2]

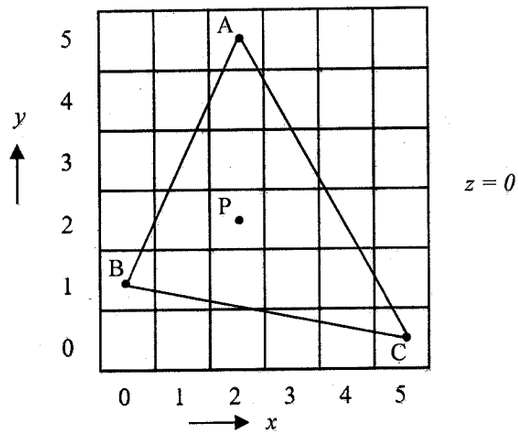
$$Par_{V,N,P_0} = \begin{bmatrix} d_1 - an_1 & -an_2 & -an_3 & ad_0 \\ -bn_1 & d_1 - bn_2 & -bn_3 & bd_0 \\ -cn_1 & -cn_2 & d_1 - cn_3 & cd_0 \\ 0 & 0 & 0 & d_1 \end{bmatrix}$$

b) Suppose you want to align vector $\bar{p} = (0, 1, 1)$ with the vector $\bar{q} = (1, 0, 1)$ by rotating around the axis $(-1, -1, 1)$. Consider the given quaternion-based rotation matrix $M_R(\theta)$, where symbols have their usual meanings. What will be the angle of rotation? (12) [CO3]

$$M_R(\theta) = \begin{bmatrix} 1 - 2b^2 - 2c^2 & 2ab - 2sc & 2ac + 2sb \\ 2ab + 2sc & 1 - 2a^2 - 2c^2 & 2bc - 2sa \\ 2ac - 2sb & 2bc + 2sa & 1 - 2a^2 - 2b^2 \end{bmatrix}$$

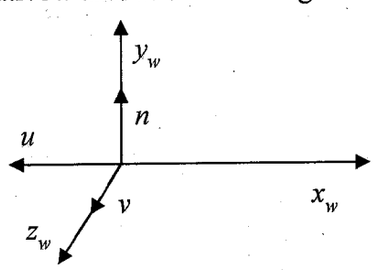
c) Derive a transformation matrix to convert a frustum shape eye-space view volume in a canonical view volume in normalized device coordinate space. (11) [CO1]

7. a) Consider a triangular fragment ABC as shown below. Let, the vertex normals are $N_A = (0, 0, 1)$, $N_B = (0, 0.2, 1)$, and $N_C = (0, -0.2, 1)$. If the light source is at $(7, 3, 4)$, viewer is at $(-6, 3, 3)$ with light color $(0.9, 0.9, 1)$, surface materials reflectivity $(0.5, 0.6, 0.4)$ and you are rendering the surface with Phong shading then determine the diffuse color at point P. (14) [CO1]



b) How can you generate soft shadows using ray tracing? Explain. (06) [CO2]

c) The current position of a camera is at $(10, 20, -15)$, looking at $(35, 40, 20)$, and head up is along $(3i + 4j - 5k)$. Find the transformation matrix that would align the camera looking direction along with positive y-axis, and the direction that points to the right of the camera along with the negative x-axis as shown in figure below. (15) [CO1]



8. a) Let a ray is defined by $\mathbf{s} = (0, 0, 0)$, $\mathbf{d} = (-1/\sqrt{14}, 3/\sqrt{14}, -2/\sqrt{14})$. (15) [CO2]

i) Determine if the ray intersects the triangle defined by the points $(-5, 1, -5)$, $(1, -1, -4)$, and $(0, 18, -7)$,

ii) Also find the transmitted ray considering the triangle as transparent surface with $\eta = 1.5$.

b) Consider following two tables showing information regarding four patches and their form factors. Determine radiosity of each patch. (15) [CO1]

Patch No	1	2	3	4
Reflectivity, ρ_i	0.2	0.5	0.6	0.4
Emissive intensity, E_i	0.8	0	0	0
Patch area	0.2	0.6	0.5	0.7

	1	2	3	4
1	1	0.6	0.6	0.3
2	–	1	0.3	0.1
3	–	–	1	0.1
4	–	–	–	1

Form Factor, F_{ij}

Approximate values up to three decimal places.

c) Given that a hemicube is defined at origin with the z axis vertical. What is the delta form factor of the hemicube pixel on the right face $x = 1$, if $(y_{min}, y_{max}, z_{min}, z_{max}) = (0.55, 0.65, 0.35, 0.45)$. (05) [CO3]